

CPCS 202 – Programming I

02 - Algorithms and Flowcharts

Objectives

- Algorithms and Flowchart
- Exercises

Algorithm – pseudocode and flowchart

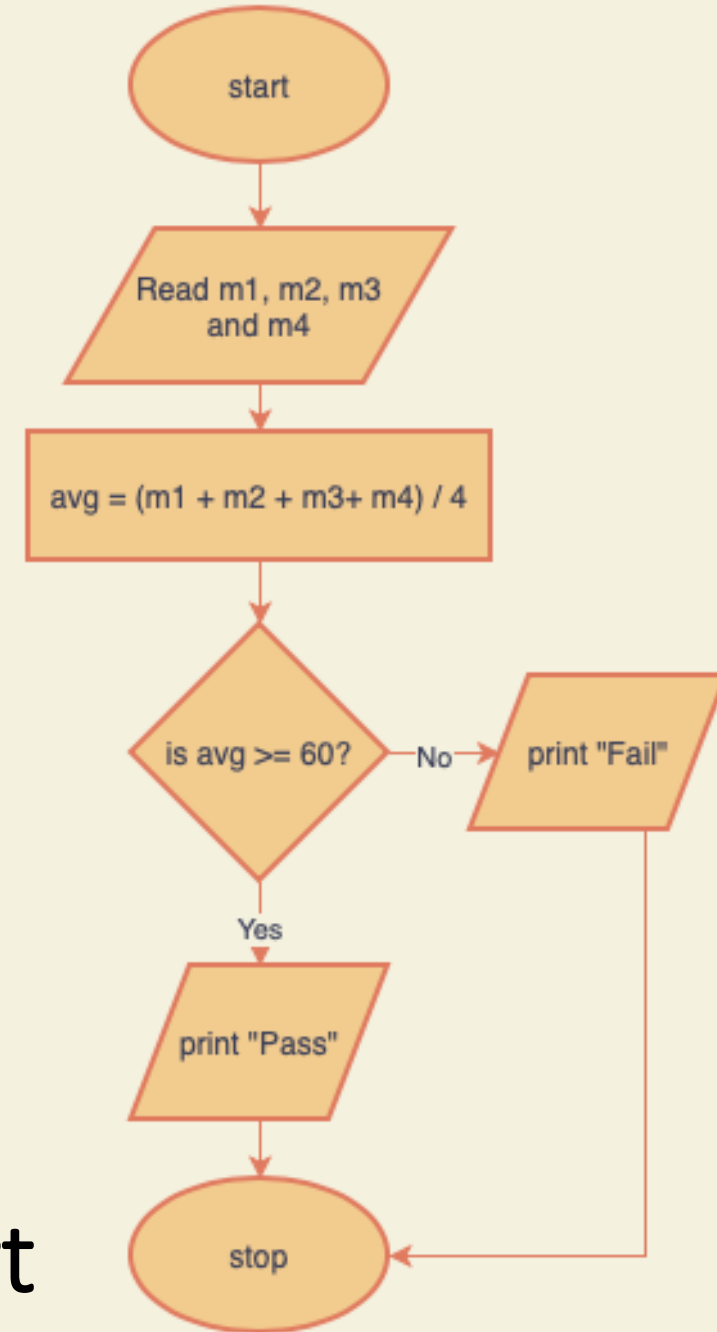
Example #1

Calculate and display a student's final quiz grade that is calculated as the average of marks obtained in four quizzes. The student is awarded a "pass" grade if his average is equal to or greater than 60. Otherwise he is awarded a "Fail" grade. Indicate whether the final quiz grade is a Pass or a Fail.

Example #1- Pseudocode

```
1: read m1, m2, m3 and m4
2: average = (m1+m2+m3+m4) / 4
3: IF average >=60
4:     print "Pass"
5: ELSE
6:     print "Fail"
7: ENDIF
```

Flowchart



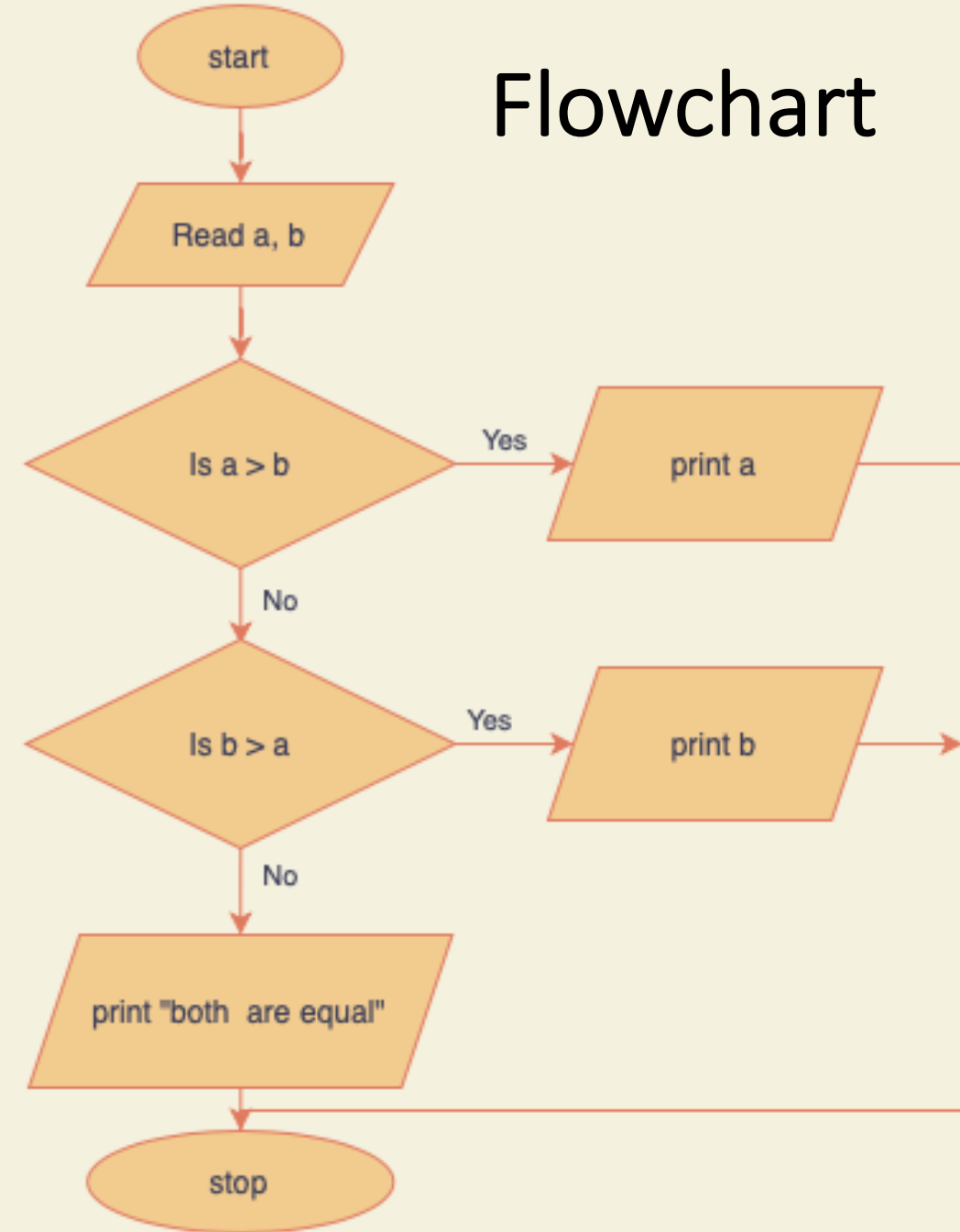
Example #2 – Larger of two numbers

Read two numbers from the user and then display the larger of the two numbers. If both are equal then display a message that both numbers are equal.

Example #2 - Pseudocode

```
1: read a, b
2: IF a > b
3:   print a
4: ELSE IF b > a
5:   print b
6: ELSE
7:   print "both are equal"
8: ENDIF
```

Flowchart



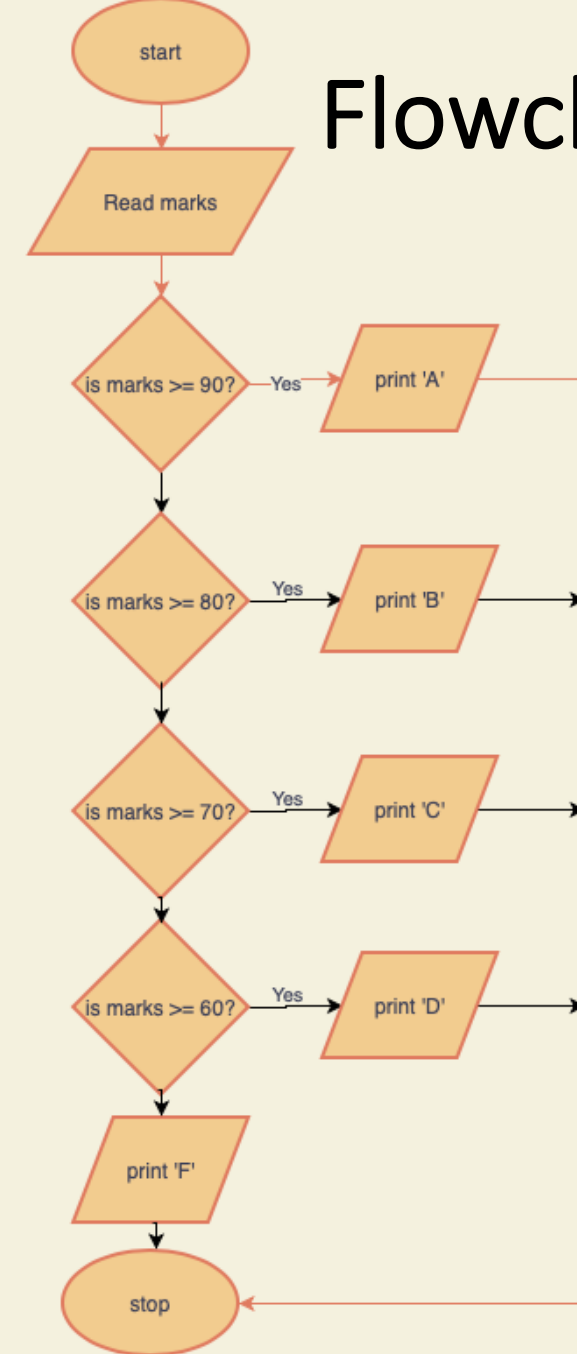
Example #3 – Grade in course

Read the total semester marks for a course from the user and then display his grade in the course (based on the KAU grade policy).

Example #2 - Pseudocode

```
1: read marks
2: IF marks >= 90
3:     print 'A'
4: ELSE IF marks >= 80
5:     print 'B'
6: ELSE IF marks >= 70
7:     print 'C'
8: ELSE IF marks >= 60
9:     print 'D'
10: ELSE
11:     print 'F'
12: ENDIF
```

Flowchart



Example #3 – Pooja and ATM

Pooja would like to withdraw X \$US from an ATM. The cash machine will only accept the transaction if X is a multiple of 5, and Pooja's account balance has enough cash to perform the withdrawal transaction (including bank charges). For each successful withdrawal, the bank charges 0.50 \$US. Calculate Pooja's account balance after an attempted transaction. (Source : <https://www.codechef.com/problems/HS08TEST>)

Sample input and output

Input	Output	Explanation
30 120.00	89.50	Successful Transaction
42 120.00	120.00	Unsuccessful transaction. Amount not multiple of 5
300 120.00	120.00	Unsuccessful transaction. Insufficient balance

Class Exercise (15 to 20 minutes)

Task description : Design the algorithm to solve the problem in Example #3 as pseudocode and as a flowchart

Work in a group of 2-3 students

Share your solutions with your instructor and your fellow students after you complete

Note : Be careful! There's a **tricky** case that you need to consider for a 100% correct solution

Example #3 – Pooja and ATM

```
1: read X, Y
2: IF X div by 5 AND X + 0.5 <= Y
3:   B = Y - X - 0.5
4:   print B // Successful
5: ELSE
6:   print Y // Not successful
7: ENDIF
```

Pseudocode

Flowchart

